Project overview

Student Profile Verification System

Automated document checks for scholarship applications. The system checks, the admin decides.

Students upload

CNIC, marksheet and degree from one simple form.

System verifies

Reads documents with OCR and cross-checks the data.

Admin approves

A human always makes the final decision.

Why we built it

The problem today

Slow manual checking

Every document is read by hand, one by one.

Missing documents

Endless back-and-forth with students.

Mismatched data

Wrong names or CNIC numbers slip through.

Scattered records

No single report per student.

Workflow

How it works, step by step

1. **Student submits the form**
   Name, email, phone and available documents. They receive a private access token.

2. **System checks everything**
   File safety, completeness score, OCR reading and data extraction.

3. **Student fixes gaps**
   The dashboard shows exactly what is missing. Only those files are uploaded.

4. **Admin reviews**
   Extracted data, warnings and a PDF dossier. Approve, reject or request changes.

5. **Export**
   CSV and ZIP of all approved students.

Completeness

Progress score: four parts, 25% each

Profile details

Validated when the form is submitted.

CNIC front

Required document.

CNIC back

Required document.

Marksheet

Required document.

Below 100% the status is INCOMPLETE. At 100% it becomes PENDING_REVIEW and goes to the admin. Degree, domicile and income proof are optional extras.

What the system reads from documents

From CNIC

CNIC number, father name, date of birth.

From marksheet

Roll no, marks, percentage, CGPA, year, board.

From degree

Degree title and institution.

Scanned photos are read with iLovePDF OCR. Text PDFs are read locally for free. Each document is read only once, so OCR credits are not wasted.

Automatic consistency checks

- CNIC number on the front matches the back
- Father name on CNIC matches the marksheet and degree
- Date of birth on CNIC matches academic documents
- Student's name appears on each document
- Optional minimum percentage or CGPA

Failed checks become warnings for the admin. The system never rejects anyone on its own, because OCR can misread genuine documents.

Roles

Who does what

Student

Submits details, uploads documents, tracks progress and fixes issues.

System

Checks files, scores completeness, reads and compares data, raises flags.

Admin

Reviews the dossier, approves or rejects, exports CSV and ZIP.

Security

How student data is protected

Access tokens

Each student sees only their own record.

Admin key

All admin actions are locked.

Private files

CNIC images are never public URLs.

Expiring links

Signed, time-limited links in CSV and dossier.

Safe uploads

File type, size and content are checked.

Audit history

Every action is logged with time.

Technology

What it is built with

FastAPI (Python)

REST API with live documentation at /docs.

SQLite

Data survives restarts.

iLovePDF API

Cloud OCR for scanned files.

ReportLab

PDF dossier that works on any server.

Headless backend: any frontend (React, mobile, web) connects through the API. Full details are in INTEGRATION_GUIDE.md.

Status

Where we are and what's next

Done	Backend API, security, OCR integration, PDF/CSV/ZIP exports, documentation
Next	End-to-end test with real CNICs and marksheets
Next	Frontend team connects student portal and admin dashboard
Next	Production server: domain, HTTPS, new keys, backups
Required	Student consent note about third-party OCR processing